UNIT - 2

Forced Vibrations of SDOF Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Steady state forced vibration, sources of excitation, impressed harmonic force, resonance, impressed force due to unbalance, motion excitation, transmissibility and isolation, performance of different types of isolators, power absorbed by viscous damping.

1. Forced Vibration - Introduction

Forced vibration : the system is subjected to an external force (or motion) that acts continuously. Response has two parts:

- *Transient (complementary function, free vibration) - dies out due to damping.*
- *Steady state (particular integral) - persists at the frequency of excitation ω, whatever the natural frequency. This is what we study.*

Sources of excitation

Source	Examples
Unbalance (rotating mass)	Turbines, motors, fans, flywheels, wheels of vehicles - force $m_0 e\omega^2$
Reciprocating motion	IC engines, compressors, pumps - inertia forces of piston/crank
Motion of support (base)	Road roughness, earthquake, vibrating foundation, ship deck
Impact / periodic force	Forging hammers, press, gear tooth errors, cam-follower, blade passing
Fluid flow / wind	Vortex shedding, flutter, wave loading, pulsating pressure
Misalignment, loose parts	Couplings, shaft bend, bearing defects

2. Impressed Harmonic Force (SDOF with damping)

Mass m, spring k, damper c, acted by $F(t) = F_0 \sin \omega t$ (F_0 = force amplitude, ω = forcing frequency).

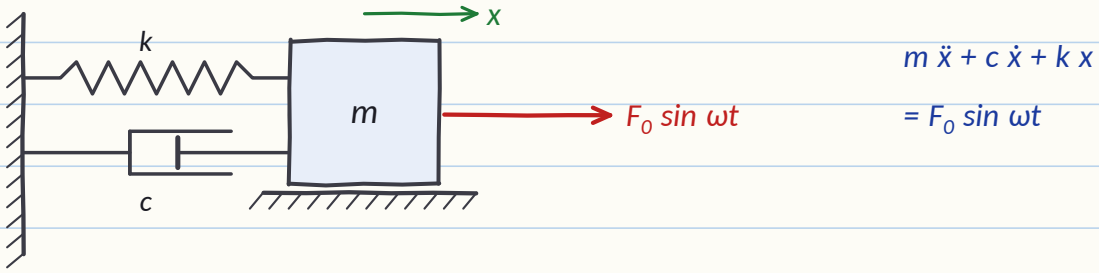


Fig 2.1 SDOF system with harmonic force

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t \quad (1)$$

Steady state solution (same frequency, lagging by phase φ): $x(t) = X \sin(\omega t - \varphi)$.

Substitute in (1) and equate the sine and cosine terms:

$$X = F_0 / \sqrt{[\,(k - m\omega^2)^2 + (c\omega)^2\,]} \qquad \varphi = \tan^{-1}[\,c\omega / (k - m\omega^2)\,] \quad (2)$$

Divide by k, use $r = \omega/\omega_n$, $\zeta = c/c_c$, static deflection $X_{st} = F_0/k$:

$$M.F. = X / X_{st} = 1 / \sqrt{[\,(1 - r^2)^2 + (2\zeta r)^2\,]}$$

$$\varphi = \tan^{-1}[\,2\zeta r / (1 - r^2)\,] \quad (0 \le \varphi \le 180°) \quad (3)$$

M.F. = magnification factor (dynamic amplification, also called dynamic magnifier).

Complete solution = transient + steady state

General solution of (1): $x(t) = x_c(t) + x_p(t)$

$$x(t) = e^{-\zeta \omega_n t}(\,A \cos \omega_d t + B \sin \omega_d t\,) + X \sin(\omega t - \varphi) \quad (2a)$$

- First term (transient) vanishes after a few time constants $1/(\zeta\omega_n)$; A and B come from initial conditions.
- Only the second term (steady state) is important for design - amplitude is independent of initial conditions.

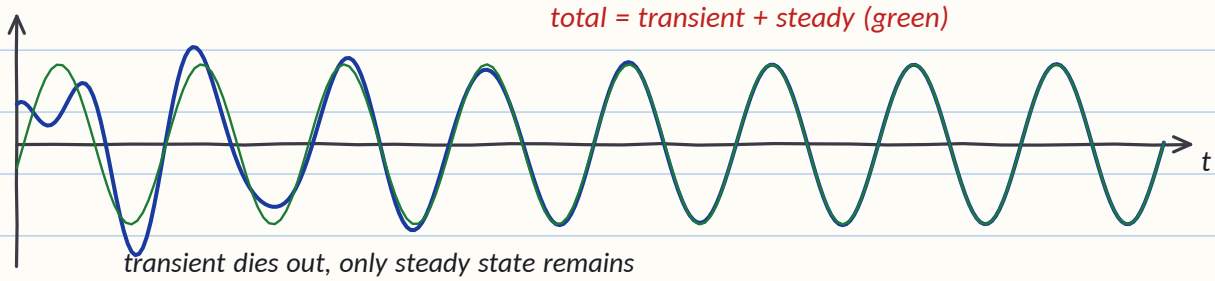


Fig 2.1a Build-up of forced response

Complex (phasor) method - frequency response function

Write $F = F_0 e^{i\omega t}$, $x = X^* e^{i\omega t}$ (X^* complex, includes phase). Then $(-m\omega^2 + ic\omega + k)\,X^* = F_0$ →

$$H(\omega) = X^*/F_0 = 1 / (k - m\omega^2 + ic\omega) \quad |H| = 1/\sqrt{[(k - m\omega^2)^2 + (c\omega)^2]}$$

$$\text{dimensionless: } H(r)\ k = 1 / (1 - r^2 + i\,2\zeta r) \qquad \textbf{(2b)}$$

The modulus gives amplitude ratio, the argument gives phase lag $\varphi = \tan^{-1}[2\zeta r/(1 - r^2)]$. This is how frequency response is measured experimentally.

Undamped forced vibration ($\zeta = 0$)

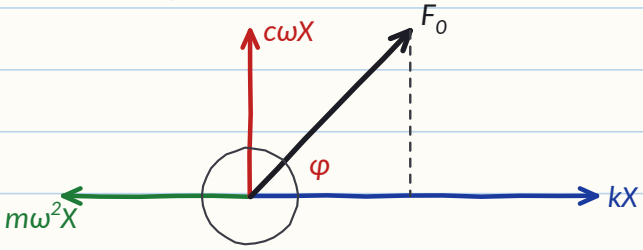
$m\ddot{x} + k\,x = F_0 \sin \omega t$. Particular solution $X = F_0/(k - m\omega^2) = X_{st}/(1 - r^2)$; response in phase ($r < 1$) or 180° out of phase ($r > 1$).

Complete solution with $x_0 = \dot{x}_0 = 0$: $x = X_{st}/(1 - r^2) \cdot (\sin \omega t - r \sin \omega_n t)$

- Beating: when ω is close to ω_n, the two frequencies add to give a slowly varying amplitude (beat frequency = $\omega_n - \omega$).
- At resonance ($\omega = \omega_n$): $x = (X_{st}/2)(\sin \omega_n t - \omega_n t \cos \omega_n t) \rightarrow$ amplitude grows linearly with time ($\sim t$); failure is a matter of time.

Vector (force polygon) representation

Four rotating vectors in equilibrium: spring force kX, damping force $c\omega X$ (leads by 90° over spring), inertia force $m\omega^2 X$ (opposite to spring), applied force F_0.



$F_0^2 = (k - m\omega^2)^2X^2 + (c\omega X)^2$

$\tan \varphi = c\omega / (k - m\omega^2)$

x lags F_0 by φ

Fig 2.2 Force polygon

3. Resonance and Frequency-Response Curves

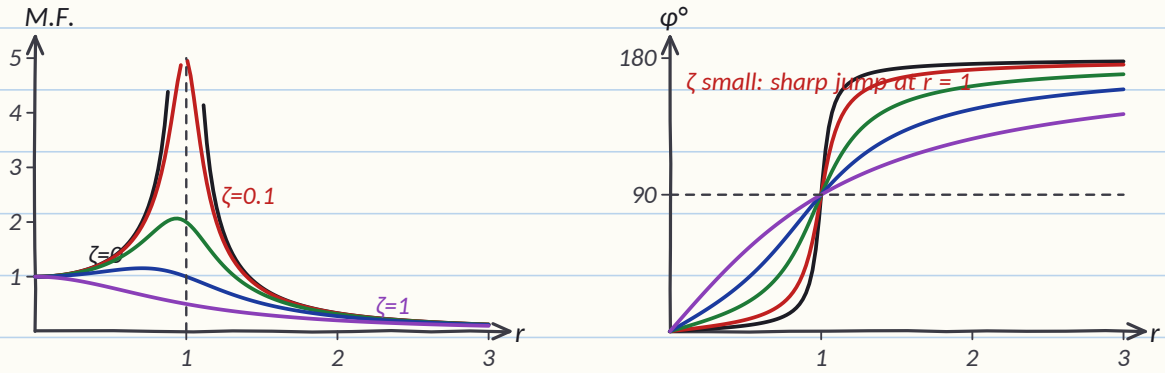


Fig 2.3 Magnification factor and phase angle vs frequency ratio r

Resonance : $r = 1$ ($\omega = \omega_n$). Amplitude is limited only by damping:

$$\text{At } r = 1: X = X_{st} / 2\zeta = F_0/(c\omega_n),\ \varphi = 90° \quad (4)$$

- Undamped ($\zeta = 0$) at resonance : $X \to \infty$ (grows linearly with time) - dangerous.
- Peak of M.F. occurs slightly left of $r = 1$: $r_{peak} = \sqrt{(1 - 2\zeta^2)}$; $M.F._{max} = 1 / (2\zeta\sqrt{(1 - \zeta^2)}) \approx 1/2\zeta$.
- No peak exists if $\zeta > 0.707$ ($1/\sqrt{2}$) - M.F. decreases monotonically from 1.

Peak amplitude - derivation

Let $D = (1 - r^2)^2 + (2\zeta r)^2$. Maximise M.F. $= D^{-1/2}$: $dD/dr = 0 \to -4r(1 - r^2) + 8\zeta^2 r = 0$

$$r_{peak}^2 = 1 - 2\zeta^2 \to r_{peak} = \sqrt{(1 - 2\zeta^2)}$$
$$M.F._{max} = 1 / (2\zeta \sqrt{(1 - \zeta^2)}) \text{ (substituting back)} \quad (4a)$$

- $r << 1$ ($\omega << \omega_n$): M.F. ≈ 1, $\varphi \approx 0$ - stiffness controlled, system follows force quasi-statically.
- $r = 1$: damping controlled ; force balances damper force; $\varphi = 90°$.
- $r >> 1$: M.F. $\approx 1/r^2 \to 0$, $\varphi \to 180°$ - mass controlled; mass hardly moves (basis of seismic instruments & isolators).

Bandwidth, half-power points and Q-factor

Frequencies ω_1, ω_2 where amplitude falls to (peak)/$\sqrt{2}$ (power falls to half). For small ζ:

$$\omega_{1,2} \approx \omega_n(1 \pm \zeta),\ Q = 1/2\zeta = \omega_n / (\omega_2 - \omega_1)$$
$$\zeta = (\omega_2 - \omega_1) / (2\omega_n) \text{ - used to measure damping experimentally} \quad (5)$$

Problem 1 (steady-state amplitude & phase)

$m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m, $F_0 = 100$ N at $\omega = 15$ rad/s. Find X and φ.

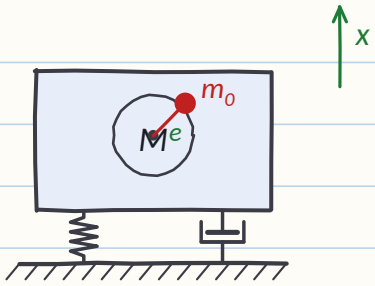
$\omega_n = 20$ rad/s, $\zeta = 40/400 = 0.1$, $r = 15/20 = 0.75$, $X_{st} = 100/4000 = 25$ mm

M.F. $= 1/\sqrt{[(1 - 0.5625)^2 + (2\times0.1\times0.75)^2]} = 1/\sqrt{(0.1914 + 0.0225)} = 2.162$

$X = 25 \times 2.162 = 54.05$ mm; $\varphi = \tan^{-1}[0.15/0.4375] = 18.9°$

4. Excitation by Rotating Unbalance

Machine total mass m (including eccentric mass m_0) on spring-damper, constrained to move vertically. Eccentric mass m_0 at eccentricity e rotates at ω.



Vertical component of centrifugal force

$F = m_0\, e\, \omega^2 \sin \omega t$

$m = M$ (total mass)

Fig 2.4 Rotating unbalance excitation

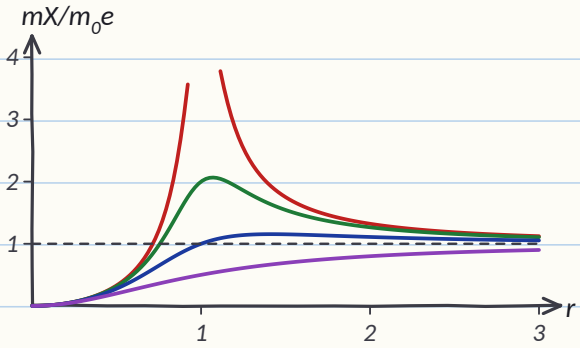
Centrifugal force vertical component $m_0\, e\, \omega^2 \sin \omega t$ - force amplitude depends on ω^2 (grows with speed).

$$m\ddot{x} + c\dot{x} + kx = m_0\, e\, \omega^2 \sin \omega t \quad (6)$$

Same as (1) with $F_0 \to m_0 e \omega^2$:

$$X = m_0\, e\, \omega^2 / \sqrt{[(k - m\omega^2)^2 + (c\omega)^2]}$$

$$m\,X / (m_0\, e) = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \quad \varphi = \tan^{-1}[2\zeta r/(1 - r^2)] \quad (7)$$



At $r = 1$: $mX/(m_0 e) = 1/2\zeta$

$r \to \infty$: $X \to m_0 e / m$

(independent of damping)

$r \to 0$: $X \to 0$

ζ=0.1 ζ=0.25 ζ=0.5 ζ=1

Fig 2.5 Response to rotating unbalance

- *Peak occurs at $r = 1/\sqrt{(1 - 2\zeta^2)}$ (slightly right of resonance); $M.F._{max} = 1 / (2\zeta\sqrt{(1 - \zeta^2)})$.*
- *Reciprocating machines: the primary inertia force $\approx m_0 e\omega^2 \cos \omega t$ - the same analysis applies (m_0 = reciprocating mass).*
- *Force transmitted to foundation: $F_T = X\sqrt{(k^2 + (c\omega)^2)} = kX\sqrt{(1 + (2\zeta r)^2)}$.*

Problem 2 (rotating unbalance)

Machine $M = 200$ kg on springs $k = 80$ kN/m, $c = 800$ N·s/m; unbalance $m_0 e = 0.4$ kg·m; speed 300 rpm. Find amplitude, phase and amplitude at resonance.

$\omega_n = \sqrt{(80000/200)} = 20$ rad/s; $\zeta = 800/(2\sqrt{(80000 \times 200)}) = 800/8000 = 0.1$; $\omega = 2\pi(300)/60 = 31.42$ rad/s; $r = 1.571$

$mX/(m_0 e) = r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} = 2.467/\sqrt{(2.153 + 0.0987)} = 2.467/1.501 = 1.644$

$X = 1.644 \times 0.4/200 = 3.29$ mm; $\varphi = 180° - \tan^{-1}(0.314/1.467) = 167.9°$

At resonance (≈ 191 rpm): $X = (m_0 e/m)/(2\zeta) = 0.002/0.2 = 10$ mm.

Problem 2a (limiting the amplitude)

For the machine of Problem 2 find the speed at which amplitude is maximum, and the maximum amplitude.

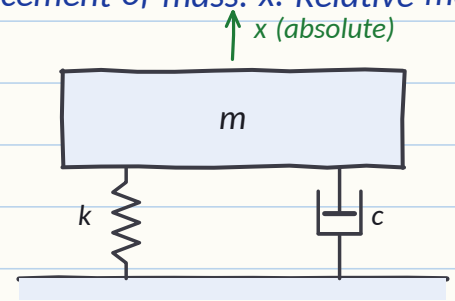
$r_{peak} = 1/\sqrt{(1 - 2\zeta^2)} = 1/\sqrt{0.98} = 1.010 \rightarrow \omega = 1.010 \times 20 = 20.2$ rad/s $\rightarrow N = 193$ rpm

$(mX/m_0 e)_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)}) = 1/(0.2 \times 0.995) = 5.025 \rightarrow X_{max} = 5.025 \times 0.002 = 10.05$ mm

Well above resonance ($r \geq 3$, i.e. $N > 570$ rpm) the amplitude approaches $m_0 e/m = 2$ mm ($r = 3$ gives 2.24 mm) - run the machine well above resonance.

5. Motion Excitation (Support / Base Excitation)

Base moves harmonically $y = Y \sin \omega t$ (road, foundation, earthquake). Absolute displacement of mass: x. Relative motion: $z = x - y$.



Eqn: $m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$

or $m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$

RHS = $Y\sqrt{(k^2 + (c\omega)^2)}\ \sin(\omega t + \alpha)$

Fig 2.6 Base-excited system

Solution $x = X \sin(\omega t - \varphi_1)$. Ratio of mass amplitude to base amplitude (displacement transmissibility):

$$X / Y = \sqrt{[1 + (2\zeta r)^2]} / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$$

$$\varphi_1 = \tan^{-1}[\ 2\zeta r^3 / (1 + (4\zeta^2 - 1)\ r^2)\ ] \quad (8)$$

Relative motion $z = x - y$ (what a vibrometer / seismic instrument measures):

$$m\ddot{z} + c\dot{z} + kz = m\omega^2 Y \sin \omega t \rightarrow Z / Y = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \quad (9)$$

- Force transmitted to base through spring + damper: $F_T = k(x-y) + c(\dot{x}-\dot{y}) = -m\ddot{x} \rightarrow$ amplitude $F_T = m\omega^2 X$.

Problem 3 (base excitation)

$m = 50$ kg, $k = 20$ kN/m, $\zeta = 0.2$; base moves with $Y = 5$ mm at $\omega = 30$ rad/s. Find X and force transmitted.

$\omega_n = \sqrt{(20000/50)} = 20$ rad/s; $r = 1.5$; $2\zeta r = 0.6$

$X/Y = \sqrt{(1 + 0.36)}/\sqrt{[(1 - 2.25)^2 + 0.36]} = 1.166/1.387 = 0.841 \rightarrow X = 4.21$ mm

$F_T = m\,\omega^2 X = 50\,(900)(0.00421) = 189$ N.

Problem 4 (vehicle on a wavy road)

A car of 1200 kg, suspension $k = 400$ kN/m, $\zeta = 0.5$, moves at 100 km/h on a road with sinusoidal profile, amplitude 50 mm, wavelength 6 m. Find the amplitude of vertical motion of the car body and relative to the wheels.

$v = 27.78$ m/s; $\omega = 2\pi v/\lambda = 2\pi(27.78)/6 = 29.09$ rad/s; $\omega_n = \sqrt{(400000/1200)} = 18.26$ rad/s; $r = 1.593$

$2\zeta r = 1.593$; $X/Y = \sqrt{(1 + 2.539)}/\sqrt{[(1 - 2.539)^2 + 2.539]} = 1.881/2.215 = 0.849$

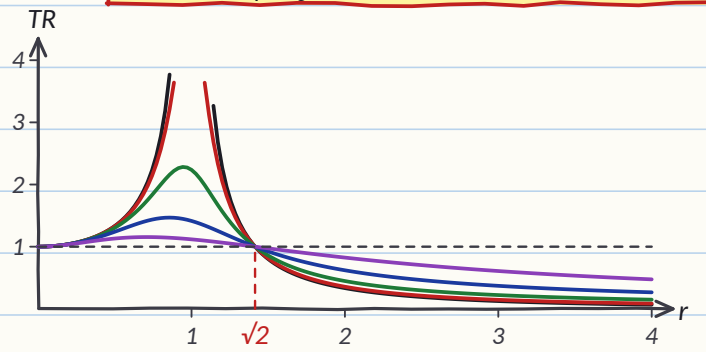
$X = 0.849 \times 50 = 42.5$ mm. Relative: $Z/Y = r^2/\sqrt{[\ldots]} = 2.539/2.215 = 1.146 \rightarrow Z = 57.3$ mm (suspension travel).

6. Transmissibility and Vibration Isolation

Vibration isolation : reducing transmission of vibration from machine to foundation (force isolation) or from foundation to equipment (motion isolation). Done by placing resilient mounts (isolators).

Transmissibility (TR) = ratio of transmitted force amplitude to applied force amplitude (or output motion to base motion). For both cases:

$$TR = F_T/F_0 = X/Y = \sqrt{[1 + (2\zeta r)^2]} \,/\, \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \qquad (10)$$



$r < \sqrt{2}$: amplification

$r > \sqrt{2}$: isolation ($TR < 1$)

all curves cross at $r = \sqrt{2}$

ζ=0 ζ=0.1 ζ=0.25

ζ=0.5 ζ=1

Fig 2.7 Transmissibility curves

- Isolation ($TR < 1$) only when $r > \sqrt{2}$, i.e. $\omega_n < \omega/\sqrt{2}$. So the isolator must be soft (low $k \rightarrow$ large static deflection).
- For $r > \sqrt{2}$, more damping increases TR (worse isolation); for $r < \sqrt{2}$ damping helps (limits resonance). Compromise: small damping ($\zeta \approx 0.05$-0.2) to pass resonance safely at start-up.

- Undamped: $TR = 1/(r^2 - 1)$ (magnitude). Isolation efficiency $\varepsilon = (1 - TR) \times 100\ \%$.

$$r = \sqrt{(1 + 1/TR)}\ \text{(undamped)};\ \delta_{st} = g/\omega_n^2;\ f_n = 15.76/\sqrt{\delta_{st}}(\text{mm})\ \text{Hz} \quad (11)$$

Problem 5 (design of isolator)

A 500 kg machine runs at 1200 rpm. Required isolation 80% (TR = 0.2); neglect damping. Find k and static deflection.

$\omega = 2\pi(1200)/60 = 125.7$ rad/s; $r = \sqrt{(1 + 1/0.2)} = \sqrt{6} = 2.449$; $\omega_n = 125.7/2.449 = 51.3$ rad/s

$k = m\,\omega_n^2 = 500(51.3)^2 = 1.316 \times 10^6$ N/m; $\delta_{st} = g/\omega_n^2 = 9.81/2632 = 3.73$ mm

(Total k; with 4 mounts each k = 329 kN/m.)

Problem 6 (force transmitted with damping)

Machine 120 kg, unbalanced force 400 N at 1800 rpm, on mounts of total $k = 1.2 \times 10^5$ N/m, $\zeta = 0.1$. Find the amplitude and force transmitted.

$\omega = 188.5$ rad/s; $\omega_n = \sqrt{(120000/120)} = 31.62$ rad/s; $r = 5.96$; $2\zeta r = 1.192$

$M.F. = 1/\sqrt{[(1 - 35.53)^2 + 1.421]} = 1/34.55 = 0.0289$; $X_{st} = 400/120000 = 3.33$ mm

$X = 3.33 \times 0.0289 = 0.096$ mm; $TR = \sqrt{(1 + 1.421)} \times 0.0289 = 0.045 \rightarrow F_T = 0.045 \times 400 = 18$ N (95.5% isolation).

Isolation vs. absorption (comparison)

Isolator (this unit)	Dynamic vibration absorber (Unit III)
Soft mount between source and base	Auxiliary spring-mass attached to the machine
Works for a wide range of speeds ($r > \sqrt{2}$)	Works at one operating frequency (tuned)
Reduces transmitted force / motion	Reduces vibration of the main mass itself

7. Performance of Different Types of Isolators

Choice depends on load, frequency range, damping needed, environment (oil, heat), cost and space.

Isolator	Characteristics / Typical use
Steel coil springs	Large static deflection → low f_n; very low damping ($\zeta \approx 0.005$); durable; good low-frequency isolation (compressors, large machines). May transmit high-frequency noise ("surge"), so add rubber pad / damper.
Rubber / neoprene mounts	Moderate damping, used in shear or compression; compact, cheap, easy to mount (engine mounts, pumps). Deteriorates with oil, heat, ageing; limited deflection.
Cork pads	Used under heavy machinery and foundations; good for high frequency; small deflection → higher f_n; damping moderate; life limited by moisture.
Felt pads	Light loads, high-frequency noise isolation, cheap; creeps & compacts with time.
Air (pneumatic) springs	Very low f_n; stiffness adjustable by pressure; self-levelling; damping through orifice. Precision equipment; costly, needs air supply.
Wire-rope / metal mesh, hydraulic mounts	High damping with moderate stiffness; used where shock, wide temperature, or both isolation and damping are needed.

- *Effective isolator needs:* low natural frequency (large δ_{st}), enough damping for resonance crossing, stability against rocking, and compatibility with environment.
- *Foundation mass:* mounting machine on large inertia block lowers amplitude and keeps the C.G. low.

Design procedure for a passive isolator

1. Find the disturbing frequency ω (operating speed) and forcing amplitude.
2. Decide the required isolation / TR (e.g. 80%, 90%).
3. Find r from TR: $r = \sqrt{(1 + 1/TR)}$ (undamped) or from curves with the chosen ζ.
4. Natural frequency $\omega_n = \omega/r$; stiffness $k = m\omega_n^2$; $\delta_{st} = mg/k$.
5. Select isolator type to give k and a small ζ; check start-up resonance crossing and rocking stability.

- *Active / semi-active isolation:* sensors + actuators vary force or damping in real time (car magnetic-ride dampers, precision tables) - effective also at low frequency where passive mounts fail.

8. Power Absorbed by Viscous Damping

Damper force $F_d = c\,\dot{x}$. With $x = X\sin(\omega t - \varphi)$, velocity $\dot{x} = \omega X\cos(\omega t - \varphi)$. Energy dissipated per cycle:

$$\Delta W = \int_0^T c\,\dot{x}\cdot\dot{x}\,dt = \pi\, c\,\omega\, X^2 \quad (12)$$

Average power = $\Delta W / T$:

$$P_{avg} = \tfrac{1}{2}\, c\,\omega^2 X^2 = \tfrac{1}{2}\, F_0\,\omega\, X\sin\varphi \quad (13)$$

- Second form: power supplied by the exciting force = F_0 × (component of velocity in phase with F_0) - at steady state equals power dissipated in damper.
- Using X from (2): $P = \tfrac{1}{2}\, c\,\omega^2 F_0^2 / [(k - m\omega^2)^2 + (c\omega)^2]$. Maximum at resonance ($\omega = \omega_n$):

$$P_{max} = F_0^2 / (2c) \quad (14)$$

- Half-power points (ω_1, ω_2) are where $P = P_{max}/2$ - hence the name; ΔW is the area of hysteresis loop in force-displacement plot (ellipse).

Problem 7 (power absorbed)

For Problem 1 ($c = 40$, $\omega = 15$ rad/s, $X = 54.05$ mm): $P = \tfrac{1}{2}\, c\,\omega^2 X^2 = 0.5(40)(225)(0.05405)^2 = 13.1$ W.

Check: $\tfrac{1}{2}\, F_0\,\omega\, X\sin\varphi = 0.5(100)(15)(0.05405)(0.324) = 13.1$ W. Energy lost per cycle $\Delta W = \pi\, c\,\omega\, X^2 = \pi(40)(15)(0.05405)^2 = 5.5$ J ($= P \times T = 13.1 \times 0.419$).

Problem 8 (damping from half-power bandwidth)

Frequency-response test: resonance peak at 20 Hz; amplitude falls to 0.707 of peak at 18.8 Hz and 21.2 Hz. Find ζ and Q.

$\zeta = (f_2 - f_1)/(2f_n) = (21.2 - 18.8)/(2\times 20) = 0.06$; $Q = 1/2\zeta = 8.33$.

Problem 9 (damping from resonance amplitude)

A machine with static deflection $X_{st} = 1.2$ mm under F_0 shows amplitude 15 mm at resonance. Find ζ and c/c_c.

$X_{res} = X_{st}/2\zeta \rightarrow \zeta = X_{st}/(2X_{res}) = 1.2/30 = 0.04$ (4% of critical damping).

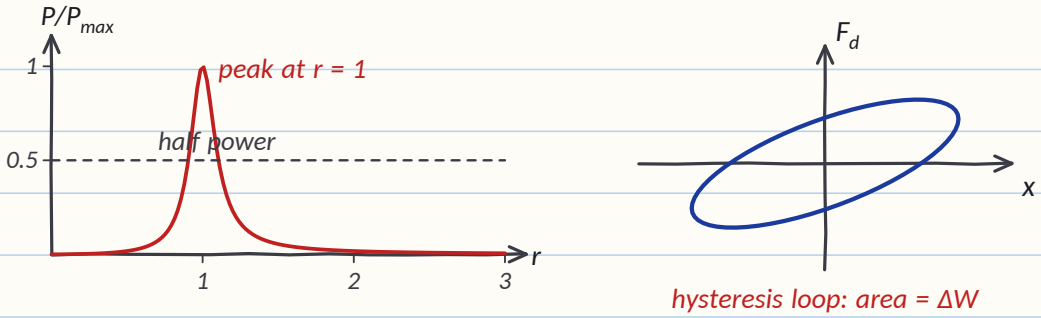


Fig 2.8 Power absorbed and damper energy loop

9. Quick Revision & Important Questions

Quantity	Formula
Force excitation	$M.F. = 1/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; $\tan\varphi = 2\zeta r/(1 - r^2)$
Resonance	$r = 1$; $X = X_{st}/2\zeta$; $\varphi = 90°$
Unbalance	$mX/(m_0 e) = r^2 M.F.$; $X \to m_0 e/m$ as $r \to \infty$
Base excitation	$X/Y = \sqrt{(1 + (2\zeta r)^2)}$ M.F.; $Z/Y = r^2 M.F.$
Transmissibility	$TR = \sqrt{(1 + (2\zeta r)^2)}$ M.F.; isolation for $r > \sqrt{2}$
Power	$P = \frac{1}{2} c \omega^2 X^2$; $P_{max} = F_0^2/2c$

10. Short Questions & Answers

1. *What is resonance?* Condition $\omega = \omega_n$ ($r = 1$) where amplitude is maximum, limited only by damping; undamped amplitude $\to \infty$.
2. *What is magnification factor?* $X/X_{st} = 1/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; ratio of dynamic to static amplitude.
3. *Why does the forced response occur at excitation frequency?* The transient dies out and the particular solution has the same frequency as the force.
4. *Define transmissibility.* Ratio of transmitted force (or motion) amplitude to the applied force (or base motion) amplitude.
5. *When is isolation effective?* Only when $r > \sqrt{2}$; then $TR < 1$. Lower the natural frequency (larger static deflection).
6. *Effect of damping beyond $r = \sqrt{2}$?* Damping increases transmissibility there, so damping should be minimal for pure isolation.
7. *Why does unbalance response tend to $m_0 e/m$ at high speed?* The mass rotates about its centre of gravity (self-centring) - amplitude independent of damping.

8. What is Q-factor? $Q = 1/2\zeta = \omega_n/(\omega_2 - \omega_1)$; measure of sharpness of resonance.
9. What phase angle at resonance? $\varphi = 90°$, independent of damping.
10. Average power absorbed by damper? $P = \frac{1}{2} c \omega^2 X^2$; at resonance $P_{max} = F_0^2/(2c)$.

Important questions: (1) derive X and φ for harmonic force; draw M.F. & phase curves (2) rotating-unbalance response (3) base excitation and transmissibility, why isolation needs $r > \sqrt{2}$ (4) isolator design problem (5) types and performance of isolators (6) power absorbed by viscous damper.